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Examining the relationships between cyberbullying, relational victimization, and family support on depressive symptoms and substance use among adolescents

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Abstract
The present study examined whether cyberbullying, relational victimization, and family support were associated with depressive symptoms and substance use among adolescents (N = 1232). A path analysis revealed that cyberbullying, relational victimization, and family support were positively associated with depressive symptoms. Also, cyberbullying and relational victimization was positively related to substance use, whereas family support was not statistically predictive of substance use. This study highlights preventive interventions for cyberbullying, relational victimization, and family support in schools and families.

Keywords
Cyberbullying, depression, substance use, relational victimization

Adolescence is characterized as a time of biological, psychological, and interpersonal growth (Patton et al., 2016). Historically, this phase involves increased parental conflict, emotional instability, and engagement in risky behavior (Viejo, Leva, Paredes, & Ortega-Ruiz, 2020). Adolescents are more likely to take risks, such as experimenting with illegal substances, than in other developmental stages (Patton et al., 2016). The variety of experiences and changes during adolescence increases the risk of developing mental health symptomology and engage in substance use. Adolescents often face several challenges without the proper understanding or strategies; they may engage in maladaptive coping to assist with mental health symptomology (Cole, Peterson, & Smith, 2018).

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Due to adolescence being characterized as a stage of growth and development (Patton et al., 2016), researchers have extensively studied several risk factors and maladaptive behaviors that are prevalent and persistent during this developmental stage (Aneshensel & Huba, 1983; Braucht, Brakarsh, Follingstad, & Berry, 1973). Although various behaviors are associated with adolescence, adolescents experience a higher rate of depressive symptoms and substance use (Aneshensel & Huba, 1983; Braucht et al., 1973; Clayborne, Varin, & Colman, 2019). Both depressive symptoms and substance use have negative implications for long-term development (Vaughan, Foshee, & Ennett, 2010). Therefore, continued research examining factors associated with adolescent depressive symptoms and substance use may assist in developing effective prevention interventions.

Depressive symptoms and substance use

Depressive symptoms may include a change in mood, behavior, cognition (Keyes, Gary, O'Malley, Hamilton, & Schulenberg, 2019), sleep disturbance, and feelings of hopelessness (Ackard, Eisenberg, & Neumark-Sztainer, 2007; Exner-Cortens, Eckenrode, & Rothman, 2013). An estimated 12.5% of adolescent individuals experience a major depressive episode within a year (Clayborne et al., 2019) and depressive symptoms (Fauth, Roth, & Brooks-Gunn., 2007; Ge, Conger, & Elder, 2001; Gutman & Eccles, 2007). Depressive symptoms increase the risk of developing a depressive disorder and increase suicide risk (Keyes et al., 2019). When exploring factors associated with an increased likelihood of depressive symptoms, negative peer engagement, such as relational aggression and familial support, play a vital role when assessing risk and protective factors for depressive symptoms (Lawrence, 2022; Vaughan et al., 2010). Many studies postulate that victimization is associated with various outcomes, including depressive symptoms (Gastic, 2008; Glew, Fan, Katon, & Rivara, 2008; Klomek, Marrocco, Kleinman, Schonfeld & Gould, 2007). Prior literature suggests that specific methods of victimization, such as cyberbullying, also increase the likelihood of developing depressive symptoms (Perren et al., 2010; Wang et al., 2011; Ybarra & Mitchell, 2004). Due to the various predictors that may increase the likelihood of experiencing depressive symptomology during adolescence, continued research examining these relationships is needed to understand methods of decreasing high prevalence rates.

Adolescent substance use, specifically alcohol, tobacco, and marijuana, is a prevalent public health concern (Johnston, O'Malley, Miech, Bachman, & Schulenberg, 2016). Substance use rates tend to increase during adolescence, with a prevalence rate estimated at 70% (Gámez-Guadix, Orue, Smith, & Calvete, 2013). The use of substances during adolescence increases the likelihood of experiencing adverse social and health outcomes (Odgers et al., 2008; Squeglia, Jacobus, & Tapert, 2009). Additionally, substance use during adolescence is associated with difficulties with interpersonal relationships (Branstetter & Furman, 2013; Miotto et al., 2013) and has long-term negative consequences that may include the development of a substance use disorder (Trucco, 2020). Research suggests that protective factors, such as family support, may decrease the likelihood of adolescent substance use (Wills, Vaccaro, & McNamara, 1992). Specifically, supportive relationships with family may increase perceived emotional support for the adolescent, which increases the likelihood of engaging in healthy coping behaviors (Cauce, Reid, Landesman, & Gonzales, 1990).

Cyberbullying

Cyberbullying is defined as using technology to repeatedly intimidate, embarrass, and be aggressive toward a target (David-Ferdon & Hertz, 2007; Grigg, 2010). Prior research estimates that

cyberbullying rates among adolescents range from 10% to 35% (Jones, Mitchell, & Finkelhor, 2013; Kowalski & Limber, 2013; Patchin & Hinduja, 2006; Slonje & Smith, 2008; Wang et al., 2009; Williams & Guerra, 2007); however, others estimate that 20%–40% of adolescents have experienced cyberbullying (Gámez-Guadix et al., 2013). Though research has established that in-person bullying is more customary, researchers also acknowledge that cyberbullying may be more harmful due to the anonymity of the aggressor and potential widespread audiences (Mitchell, Jones, Turner, Shattuck, & Wolak, 2016). Cyberbullying is associated with several adverse effects, including depressive symptoms (Fredstrom, Adams, & Gilman, 2011) and substance use (Gámez-Guadix et al., 2013). Several studies found that cyberbullied adolescents also experienced more depressive symptoms (Perren et al., 2010; Wang et al., 2011; Ybarra & Mitchell, 2004). Regarding substance use specifically, several studies have associated being a victim of cyberbullying with increased substance use (Hinduja & Patchin, 2008; Gámez-Guadix et al., 2013; Smith et al., 2008).

Relational victimization

Relational victimization is described as exposure to behaviors that damage relationships or one's social reputation, such as exclusion, manipulation, and rumor spreading (Crick & Grotpeter, 1996). Relational victimization predicts maladaptive social-cognitive processes and heightened emotions (Rudolph, Troop-Gordon, & Flynn, 2009). Prior studies indicate that victimization experiences have been shown to occur with substance use during adolescence (Kuntsche & Gmel, 2004; Sullivan, Farrell, & Kliewer, 2006; Tharp-Taylor, Haviland, & D'Amico, 2009; Weiss, Mouttapa, Cen, Johnson, & Unger, 2011). Relational victimization may also lead to internalizing problems, such as depressive symptoms and an increased desire to respond negatively to others, leading to externalizing behaviors (Crick et al., 2001). Prior research has primarily focused on physical and verbal victimization (Ostrov & Kamper, 2015). Thus, further research is needed regarding relational victimization.

Family support

During adolescence, the relationship with the family unit may have implications for development across the lifespan (Chen, Brody, & Miller, 2017). Family support, which typically involves parental figures, is a vital relationship that ultimately impacts subsequent engagement (Collins & Laursen, 2004; Smetana, Campione-Barr, & Metzger, 2006). Family support within the home often serves as a protective factor against maladjustment and depressive symptoms (Pettit, Roberts, Lewinsohn, Seeley, & Yaroslavsky, 2011). Specifically, access to familial support assists adolescents in effectively coping with challenging situations, which ultimately reduces the risk of experiencing depressive symptoms (Rosenfeld, Richman, Bowen, & Wynns, 2006).

Although family support during adolescence can be associated with more adaptive behaviors and decreased mental health symptomology, family support is not the only support or system that influences adolescent development and behavior (Bronfenbrenner, 1986). Research suggests that three main groups, family, friends, and teachers, ultimately impact adolescent development (United States Department of Health & Human Services, 2016). Based on the different relationships that tend to be influential during adolescence, it is often unclear which influence will have the most significant impact on adolescent mental health. Some studies have even suggested that the over-involvement of a family during adolescence may increase risk factors (Anderson & Henry, 1994). Therefore, further research is needed to better understand this predictive factor due to inconsistent

findings regarding the role of family and how that may be associated with depressive symptoms and substance use.

Current study

The current study examined whether family support, relational victimization, and cyberbullying are associated with depressive symptoms and substance use among adolescents. Based on prior literature, cyberbullying was hypothesized to be positively associated with depressive symptoms and substance use. Second, relational victimization would be positively related to depressive symptoms and substance use. Finally, the association between family support and substance use was examined.

Method

Participants were drawn from the Illinois Study of Bullying and Sexual Violence (Espelage, Low, Anderson, & De La Rue, 2014). Participants were middle school students (Grades 5–7); 49.8% female; 51 African American, 34% White, 3% Hispanic, 3% Asian, and 9% identified as other. Participants were 10–15 years of age (M = 13.9); SD = 1.05). The original purpose of this longitudinal study was to examine the degree to which adolescents' level of violence within the home, frequency of bullying, fighting perpetration, and alcohol/drug use influence their functioning in three waves of data. Due to using a cross-sectional design, the current study only used one wave of the data.

Procedure

Parenteral consent. First, a waiver of parental consent document was sent to the university institutional review board and the school district's administration. Upon agreement, letters were sent to all parents of students enrolled in the school. The letters informed them about the risk and purpose of the study. In addition, several meetings were held to inform parents about the study within the community. Parents were asked to sign the form and return it if they were willing to participate. At the beginning of data collection, students whose parents declined to participate were removed from the room, and students were told not to complete the survey if their parents never returned the consent form. Then students were asked to assent to the study, and those who declined were also removed from the room. A 95% participation rate was achieved.

Survey administration

Six trained research assistants and a faculty member disseminated surveys to students for data collection. Research assistants administered individual measures to classes ranging from 10 to 25 students. Research assistants ensured that students sat far enough from each other to increase confidentiality. The survey took an average of 40 minutes to complete.

Measures

Depression. Depression was measured using the six-item Orpinas Modified Depression Scale (Orpinas, 1993). This scale measures the degree to which individuals experience depressive symptoms over 30 days. An example item is "Did you feel hopeless about your future" measured on a 5-point Likert scale ranging from 1 (never) to 5 (almost always). Positively worded items were

Table 1. Correlations.

Variable	1	2	3	4	5
1. Family-S	—				
2. Cyberbullying	.03	—			
3. Relational-V	.04***	.31**	—		
4. Substance	−.02	.14***	.12***		
5. Depressive-S	.11***	.20***	.29***	.17***	—

Note. Family S = Family support; Cyberbullying = cyberbullying victimization; Relational victimization = relational victimization; Depressive-S = depressive symptoms.
*p < .05, **p < .01, ***p < .001.

reverse-scored, and items were combined with an alpha of (α = .82). Higher scores indicate more endorsement of depressive symptoms.

Substance use. Substance use was measured using the eight-item Problem Behavior Frequency Scale (Farrell, Kung, White, & Valois, 2000). The Problem Behavior Frequency scale measures substance use and frequency in the past year. Example items are "smoked marijuana" or "drunk alcohol" measured on a 5-point Likert scale ranging from 1 (never) to 5 (10 or more times). Items were averaged, which created a single item with an alpha of (α = .79). Higher scores indicate an increased frequency of substance use.

Family support. Family support measurement consisted of 13 items. Example items are "we express feelings to each other within the family" and "In our family, people are accepted for who they are." Items were averaged together, which created one item with an alpha of (α = .84). Higher scores represent increased family support.

Relational victimization. Relational victimization was measured using the five-item relational aggression victimization scale (Crick & Grotpeter, 1996). This scale measures the extent to which individuals were victims of social aggression meant to damage or destroy another person's reputation and social relationships. An example item is, "somebody spread a negative rumor about me," measured on a 5-point Likert scale ranging from 1 (never) to 5 (7 or more times). Items were averaged together, which created one item with an alpha of (α = .84). Higher scores indicate greater victimization (Table 1).

Cyberbullying victimization. Cyberbullying victimization was measured using the revised Olweus Bully/Victim Questionnaire (Solberg & Olweus, 2003). This scale measures the degree to which individuals were bullied physically or virtually (internet). Example items are "somebody made a threatening and aggressive comment about me online" and "someone spread rumors about me online, whether they are true or not." Items were averaged together, which created one item with an alpha of (α = .75). Higher scores indicate greater cyber victimization.

Results

Preliminary analyses were conducted first to examine normality skew and multicollinearity. Relational victimization had a univariate skew (i.e., skew = 1.51). Thus, a square root transformation

Table 2. Mean, standard deviation, skew, reliabilities.

Variable	Mean	*SD*	Skew	Reliabilities
Family support	2.60	.388	.89	α = 84
Cyberbullying	−.192	.291	.59	α = .75
Relational-V	−.27	.364	.93	α = .84
Depressive-S	1.53	.773	.24	α = .82
Substance use	.027	.073	.41	α = .79

was conducted. As a result, the relational victimization skew was at an acceptable level (i.e., skew = .9.31). All the other variables were within the acceptable skew range. In addition, there was no evidence of multicollinearity, and missing data were handled with multiple imputations (Table 2).

Path analysis results

To examine whether family support, relational victimization, and cyberbullying were associated with depressive symptoms and substance use, a structural equation model was conducted. First, to examine the reliability and validity of the proposed model, model fit indices were examined. The proposed model suggested an acceptable fit index: X^2 (14, 1, 579) = 22.469, $p < .001$, $X^2/df = 3.224$, comparative fit index (CFI) = .95, Tucker Lewis Index (TLI) = .97, root mean square error of approximation (RMSEA) = .06. The structural relationship among variables suggests that cyberbullying victimization (β = .26, $p < .001$), relational victimization (β = .50, $p < .001$), and family closeness (β = .19, $p < .001$) were all positively related to depressive symptoms. Then, cyberbullying victimization (β = .02, $p < .001$) and relational victimization (β = .01, $p < .001$) were both positively related to substance use. However, family support (β = −.07, $p = .16$) was not statistically predictive of substance use (Figure 1).

Discussion

The purpose of this study was to test whether cyberbullying, relational victimization, and family support relate to depressive symptoms and substance use. Based on prior literature, this study posited three hypotheses. Hypothesis one stated that cyberbullying would be positively associated with depressive symptoms and substance use (Hypothesis one supported). Results indicated that adolescents who were victims of cyberbullying were more likely to develop depressive symptoms and use illicit substances. One possible explanation for this supported hypothesis is that exposure to cyberbullying could increase feelings of hopelessness and sadness. The increased feelings of hopelessness and sadness could ultimately lead to unhealthy coping, such as substance use. Additionally, adolescents that experienced cyberbullying may have been more likely to internalize these experiences, which also increased depressive symptoms.

Hypothesis two stated that relational victimization would positively relate to depressive symptoms and substance use (Hypothesis two supported). More specifically, we found that relationally victimized adolescents were more likely to develop depressive symptoms and use substances. These findings are consistent with prior literature, which suggested that relational victimization was associated with endorsing depressive symptoms and substance use among adolescents (Desjardins & Leadbeater, 2011; Sinclair et al., 2012; Zimmer-Gembeck, Trevaskis, Nesdale, & Downey, 2014).

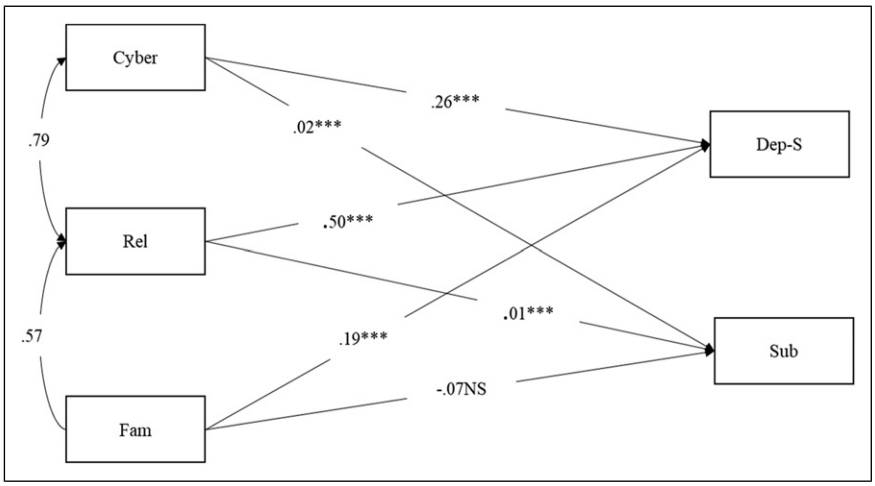


Figure 1. *Note*. Path analysis with cyber bullying victimization, relational victimization, and family support as predictors of depressive symptoms and substance use.

Results pertaining to family support suggested that although prior literature indicated that family support often decreases the likelihood for adolescents to develop depressive symptoms (Pettit et al., 2011), in the case of the current study, family support positively predicted depressive symptoms. A possible explanation for this finding is that, in some instances, perceived family support does not decrease the degree to which adolescents would endorse several depressive symptoms such as anhedonia and intense sadness (Levens, Elrahal, & Sagui, 2016). Another explanation may be related to prior literature, which indicates that a lack of secure parental attachment is often associated with depressive symptoms (Agerup, Lydersen, Wallander, & Sund, 2015; Constantine, 2006). Drawing from this literature, adolescents in this sample may not have been securely attached to their parents, relating to increased depressive symptoms. Lastly, although adolescents perceive their family as supportive, they may be less likely to discuss daily stressors, increasing their likelihood of developing depressive symptoms. Regarding family support and substance use, contrary to prior literature (Sullivan et al., 2006), family support did not predict substance use. We found that family support was not statistically predictive of substance use.

Prevention Interventions and Implications

Our results suggest that cyberbullying and relational victimization are associated with depressive symptoms and the use of illicit substances. There are several methods to disrupt these patterns. First, parents could increase monitoring behaviors of electronic communication and restrict the frequency of internet usage if adequate supervision is not possible (Ang, 2015; Baldry, Sorrentino, & Farrington, 2019; Helfrich, Doty, Su, Yourell, & Gabrielli, 2020). Second, parents and other adults may assist in increasing media literacy on how to disconnect from chat rooms or other media outlets and report cyberbullying when they suspect or experience cyberbullying. Pertaining to relational victimization, due to several studies suggesting that students are often less likely to report relational bullying experiences due to shame and feeling embarrassed (Rigby & Bagshaw, 2003; Unnever & Cornell, 2004), school counselors should implement anti-bullying interventions,

specifically to improve self-esteem, effective communication, and bystander interventions. Prior literature has revealed that anti-bullying interventions with these components often reduce relational bullying or victimization (Jacobsen & Bauman, 2007; Studer & Mynatt, 2015).

Additionally, school administrators may decrease relational victimization by implementing severe consequences to relational aggressive students to decrease the likelihood of the behavior being reinforced. Finally, schools may reduce relational victimization and cyberbullying by engaging students in psychoeducation about the effects of cyberbullying and relational victimization to create awareness and enhance empathy. The current study adds to the literature by raising further awareness of cyberbullying and relational aggression concerns among adolescents.

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